

Mapping Ghana's CORS Network: A Web-Based GIS Approach for Surveyors and GIS Professionals

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May 2026

Abstract

This article presents a practical web-based Geographic Information System (GIS) application for visualizing and Ghana's Continuous Operating Reference Station (CORS) network. The application, developed using open-source technologies including QGIS [3] and Leaflet [4], provides surveyors and GIS professionals with an intuitive tool for locating the nearest CORS, filtering by region, and accessing station information from any device with a browser. The methodology demonstrates how desktop GIS capabilities can be extended to the web environment through the qgis2web plugin [2] and custom JavaScript enhancements. The resulting mobile-responsive application features location-based services, real-time distance calculations, and a "Find Nearest CORS" functionality that enables users to identify the closest station from their current location or any user-placed point on the map. This article provides valuable insights for surveying professionals seeking to leverage geodetic infrastructure data through cost-effective, accessible web technologies.

Keywords: Continuous Operating Reference Station (CORS), GNSS, Geographic Information Systems (GIS), Web Mapping, QGIS, qgis2web, Geodetic Infrastructure, Ghana, Location-Based Services, Spatial Analysis, Surveying, Cadastral Mapping, Open Source GIS, Web GIS, Interactive Mapping, Geospatial Data, Web Development, Mobile-Responsive Applications

Related Sustainable Development Goals: SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 17 (Partnerships for the Goals).

Abbreviations and Acronyms

Table 1: Abbreviations and Acronyms

Abbreviation	Full Form
CORS	Continuous Operating Reference Station
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
GWCL	Ghana Water Company Limited
LC	Lands Commission
PWA	Progressive Web Application
QGIS	Quantum GIS
SMD	Survey and Mapping Division
WGS 84	World Geodetic System 1984

1 Introduction

The establishment of a robust geodetic infrastructure is fundamental to modern surveying, mapping, and navigation applications. Continuous Operating Reference Stations (CORS) form the backbone of this infrastructure, providing real-time GNSS corrections that enable high-precision positioning for surveying, mapping, and navigation [1].

In Ghana, the Survey and Mapping Division (SMD) of the Ghana Lands Commission has established CORS across the country to support infrastructure development, cadastral surveying, and environmental monitoring [7]. Recent developments have seen the Lands Commission deploy an effective Continuous Operating Reference System across the nation [8], with collaborative efforts between the Lands Commission, LiSAG, and GMX launching a nationwide GNSS CORS network observation [9]. This initiative represents a significant public-private partnership in

Ghana's geospatial sector [10]. However, despite the availability of this valuable geodetic data, accessibility remains a challenge for many surveyors and GIS professionals in the field. Traditional GIS software requires specialized training and is not always practical for field use.

This article presents a practical solution: an interactive web-based GIS application that makes Ghana's CORS network accessible from any device with a browser. The application is designed to address the specific needs of surveyors, GIS professionals, and field technicians working across Ghana.

2 The Ghana CORS Network

The CORS network in Ghana comprises stations strategically distributed across the country, with high-precision coordinates provided in decimal degrees. As reported by the Ghana News Agency [7], the Lands Commission has launched a digital geospatial data system that enhances access to this critical infrastructure. These stations are located at key institutions notably among them including BRRI (Kumasi), KNUST (Kumasi), various GWCL and LISAG facilities across the entire country. Table 2 summarizes the CORS included in the dataset whilst figure 1 shows the CORS network in Kilometres.

Table 2: Core details of a CORS

Region	CORS ID (Full)	Location
ASHANTI REGION	KNUST CORS/2024/1	KNUST, KUMASI
ASHANTI REGION	CORS/2025/1	GWCL, ADUM - KUMASI
ASHANTI REGION	CORS/2025/2	GWCL, OBUASI
ASHANTI REGION	CORS/2020/3	ADUM - KUMASI - LISAG
ASHANTI REGION	CORS/2025/4	GWCL, KONONGO
ASHANTI REGION	CORS/2025/5	GWCL, ASANTE MAMPONG

These stations serve multiple purposes:

- Geodetic surveying and mapping

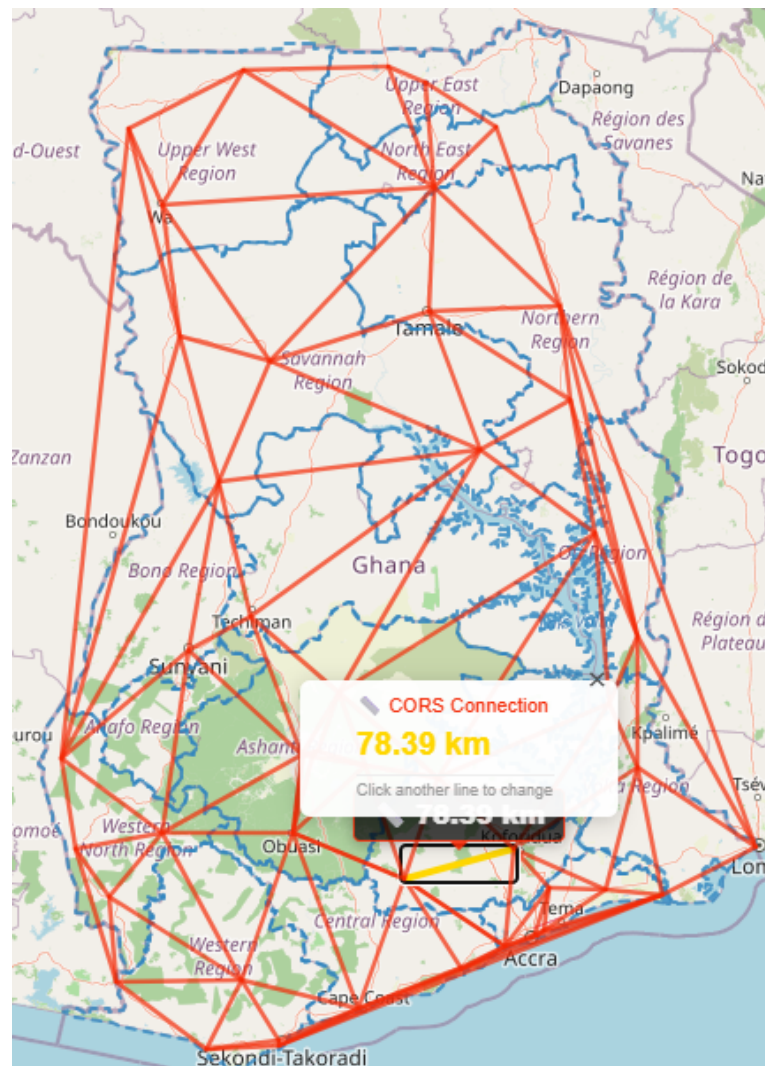


Figure 1: The CORS Network in Kilometres

- Infrastructure development
- Cadastral surveying
- Environmental monitoring
- Navigation applications

3 The Challenge: Accessibility vs. Data Availability

While the CORS network exists, several challenges limit its effective use notably among them is inability to merge different day observations to connect to a regional number :

3.1 Technical Barriers

Traditional GIS software requires specialized training, licensing, and often runs only on desktop computers. This creates a barrier for field technicians and surveyors who need quick access to station information.

3.2 Field Accessibility

Surveyors in the field need immediate information about station locations, distances, and status. They cannot always access desktop GIS software or have internet connectivity for complex systems.

3.3 Information Discovery

Knowing which CORS is closest to a project site is crucial for planning fieldwork and ensuring GNSS accuracy. However, identifying the nearest station requires spatial analysis that is not always readily available.

3.4 Data Visualization

Spatial data in tabular form is difficult to interpret. Visualizing CORS on a map provides intuitive understanding of the network's spatial distribution.

4 The Solution: A Web-Based GIS Application

To address these challenges, we developed an interactive web-based GIS application that provides:

4.1 Interactive Map Visualization

The application displays CORS on an interactive map using OpenStreetMap and satellite imagery basemaps. Key features include:

- Custom markers for CORS stations
- Station labels for easy identification
- Network lines showing spatial relationships
- Administrative region boundaries for context

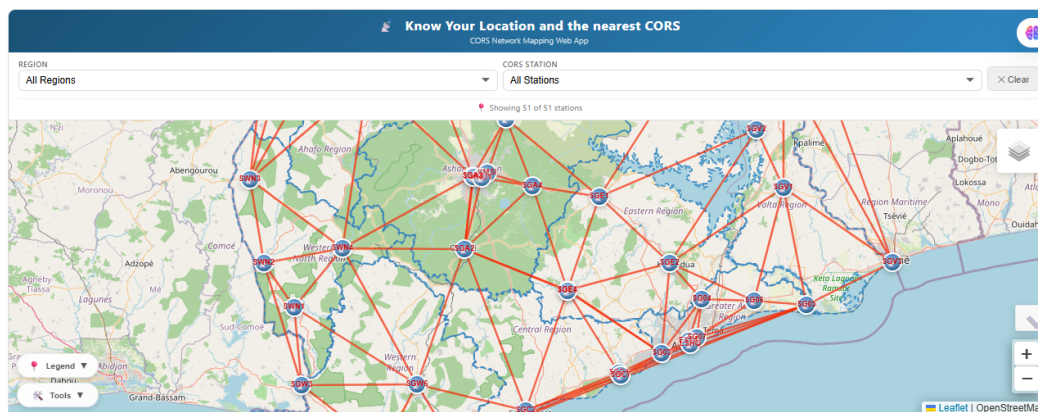


Figure 2: Web map interface showing CORS and network connections

4.2 Location-Based Services

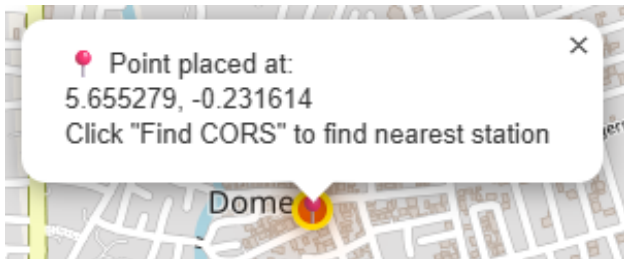
4.2.1 My Location

Users can share their location with the application. The map zooms to their position and shows their location relative to CORS. This helps surveyors understand which stations are nearby for fieldwork planning.

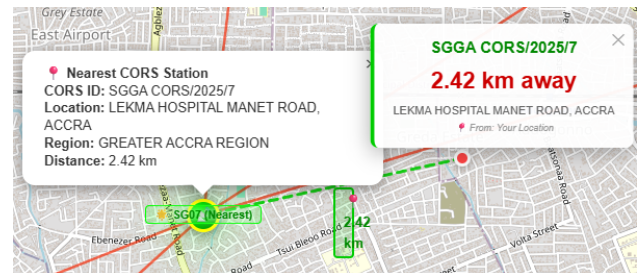
4.2.2 Find Nearest CORS

This feature allows users to find the nearest CORS from their current location or from any point they place on the map. The application:

- Calculates distances using the Haversine formula
- Highlights the nearest station
- Draws a line between the user's point and the station
- Shows station details and distance information



(a) Point placed at Dome to find nearest CORS



(b) Nearest CORS found

Figure 3: Determining the nearest CORS from a placed point: (a) User places a point at Dome, (b) The application identifies and highlights the nearest CORS

4.3 Filtering and Query Capabilities

4.3.1 Region Filtering

Users can filter CORS by administrative region, making it easier to focus on areas of interest. This is particularly useful when working in a specific region of Ghana. This is demonstrated in figure 4

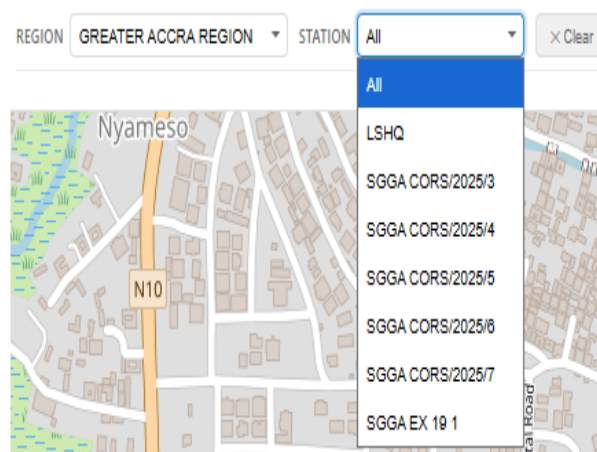


Figure 4: The Regional and station filtering of the application

4.3.2 Station Selection

Users can select individual stations from a dropdown list of both region and CORS IDs. The application zooms to the selected station and opens its information popup as shown in figure 5.

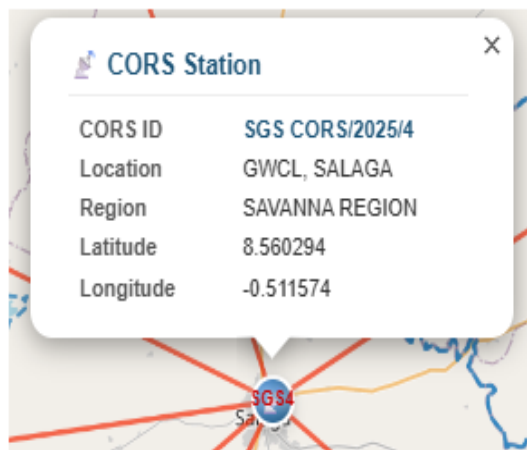


Figure 5: CORS basic information

4.4 Measurement Tools

The application includes distance measurement tools that allow users to:

- Measure distances in meters and kilometers
- Calculate areas in square meters, hectares, and acres
- Visualize measurements on the map

4.5 Mobile-Responsive Design

The application is designed to work on all devices:

- **Desktop:** Full-featured interface with all tools
- **Tablet:** Optimized layout with appropriate controls
- **Mobile:** Touch-friendly interface with compact design

5 Technical Implementation

The application was developed using a combination of open-source technologies, providing a cost-effective and maintainable solution.

5.1 Data Management (QGIS)

The CORS data was managed using QGIS [3], a free and open-source GIS software. This allowed for:

- Importing and validating station coordinates
- Managing attributes (station codes, locations, regions)
- Visualizing the spatial distribution
- Exporting data to web-compatible formats

5.2 Web Development (qgis2web)

The qgis2web plugin [2] for QGIS was used to export the project to an interactive web map. The plugin automatically generates:

- HTML structure
- CSS styling
- Leaflet JavaScript implementation
- Data files in GeoJSON format

5.3 Custom Enhancements

The qgis2web output was enhanced with custom JavaScript to add:

- Geolocation services

- Find Nearest CORS functionality
- Region filtering
- Distance calculations
- PWA capabilities [5]

5.4 Hosting (GitHub Pages)

The application is hosted on GitHub Pages, a free hosting service that provides:

- Reliable availability
- Fast loading times
- HTTPS security
- Version control integration

5.5 PWA Implementation

Progressive Web Application (PWA) features [5] were added to enable:

- Offline functionality through service worker caching
- Home screen installation on mobile devices
- Full-screen application experience
- Fast loading through cached resources

6 Feature Walkthrough

6.1 Accessing the Application

The application is accessible online at:

<https://geosolutions1-dev.github.io/cors-network-mapping/>

6.2 Main Interface

The main interface consists of:

1. **Map View:** Central area displaying the interactive map
2. **Tools Panel:** Collapsible panel with all application tools
3. **Legend Panel:** Collapsible legend explaining map symbols
4. **Status Bar:** Real-time feedback for user actions

6.3 Finding Your Location

To find your current location:

1. Click the “My Location” button
2. Grant location permission when prompted
3. The map zooms to your location
4. A marker appears at your position

6.4 Finding the Nearest CORS

To find the nearest CORS:

Option A: From Your Location

1. Click “My Location” to share your location
2. Click “Find CORS” in the tools panel
3. The application finds and highlights the nearest station

4. Distance and station details are displayed

Option B: From a Placed Point

1. Click “Place Point” to enter placement mode
2. Click anywhere on the map to place a point
3. Click “Find CORS” to find the nearest station from that point
4. The application highlights the nearest station and shows distance

6.5 Filtering by Region

To filter stations by region:

1. Select a region from the “Region” dropdown
2. The map updates to show only stations in that region
3. The station dropdown updates to show available stations

6.6 Selecting a Station

To select a specific station:

1. Choose a station from the “CORS” dropdown
2. The map zooms to the station
3. The station is highlighted and its popup opens

6.7 Measuring Distance

To measure distances:

1. Click the measurement tool button

2. Click on the map to set points
3. View distances in meters, kilometers, hectares, and acres
4. Clear measurements when done

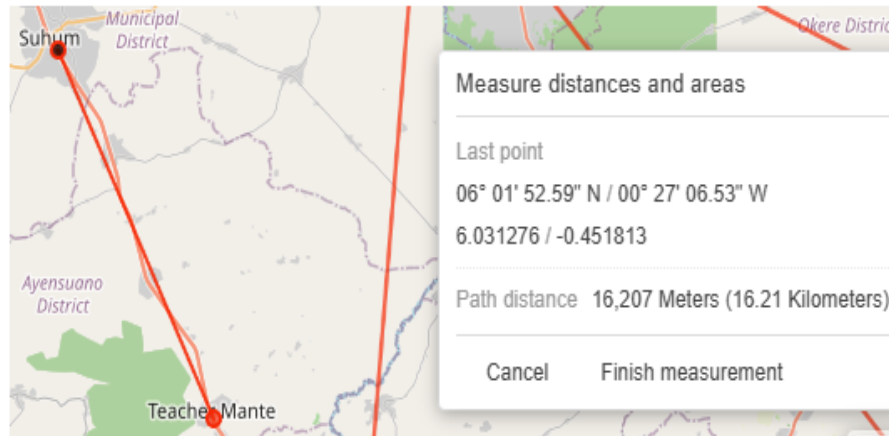


Figure 6: Distance measure tool

A distance between Teacher Mantey to Suhum has been measured to be 16.21 Kilometres as shown in figure 6

7 Benefits for Surveyors and GIS Professionals

7.1 Field Planning

Surveyors can use the application to:

- Identify the nearest CORS before heading to the field
- Plan efficient routes to stations
- Understand the spatial distribution of the network
- Verify station locations and details

7.2 Quality Assurance

The application supports quality assurance by:

- Providing accurate station coordinates in graticles
- Showing network relationships
- Enabling quick verification of station information

7.3 Training and Education

The application serves as a training tool for:

- Surveying students learning about CORS networks
- GIS professionals understanding geodetic infrastructure
- Field technicians needing station information

7.4 Accessibility

The web-based nature of the application provides:

- Access from any device with a browser
- No software installation required
- No licensing costs
- Mobile access for field use

7.5 Cost-Effectiveness

The use of open-source technologies means:

- No software licensing fees
- Free hosting on GitHub Pages
- Free GIS software (QGIS) [3]
- Free web mapping libraries (Leaflet) [4]

8 Limitations and Future Improvements

8.1 Current Limitations

- **Offline Capability:** Limited offline functionality
- **Real-Time Data:** No real-time GNSS data integration
- **Network Coverage:** Gaps in coverage in some regions
- **Station Status:** No indication of station operational status

8.2 Future Improvements

- **Real-Time Integration:** Incorporate live GNSS data feeds
- **Enhanced Offline Support:** Full offline functionality for field work
- **Status Indicators:** Show operational status of stations
- **Advanced Routing:** Provide directions to stations
- **Expanded Network:** Include additional stations as they become available

9 How to Install on Mobile (Android)

The application can be installed as a standalone app on Android devices:

9.1 Step 1: Open in Chrome

Open Chrome browser on your Android device and navigate to the application URL.

9.2 Step 2: Tap the Menu

Tap the three dots (⋮) in the top-right corner.

9.3 Step 3: Select “Install App”

Look for “Install App” or “Add to Home Screen” in the menu.

9.4 Step 4: Confirm Installation

Tap “Install” when prompted.

9.5 Step 5: Launch the App

The app icon will appear on your home screen. Tap to launch.

10 Conclusion

The web-based CORS application presented in this article addresses a critical need in the surveying and GIS community: accessible, user-friendly access to geodetic infrastructure information. As highlighted by the World Geospatial Industry Council [10], Ghana’s nationwide CORS network represents an exemplar geospatial public-private partnership that demonstrates the value of collaborative approaches to geodetic infrastructure development. By combining open-source GIS tools [3] with modern web technologies [4, 2], we have created a solution that is:

- **Accessible:** Available from any device with a browser
- **Cost-Effective:** Built with free, open-source technologies
- **User-Friendly:** Designed for surveyors and GIS professionals
- **Mobile-Ready:** Works on smartphones and tablets
- **Scalable:** Can be expanded as the network grows

The recent launch of Ghana’s digital geospatial data system [7] and the deployment of an effective Continuous Operating Reference System [8] underscore the importance of making such infrastructure accessible to professionals in the field. The collaborative efforts between the

Lands Commission, LiSAG, and GMX [9] demonstrate the commitment to advancing geospatial capabilities in Ghana.

For surveyors, GIS professionals, and field technicians working across Ghana, this application provides a valuable tool for:

- Identifying the nearest CORS
- Planning field work
- Verifying station information
- Understanding the spatial distribution of the network

Try the Application

The web application can be assessed via the link below

<https://geosolutions1-dev.github.io/cors-network-mapping/>

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Appendix: Key Features

Table 3: Key Features of the CORS Web Application

Feature	Description
Interactive Map	View CORS and network lines
My Location	Find your current location on the map
Find Nearest CORS	Locate the closest CORS
Place Point	Place a point and find the nearest station
Region Filtering	Filter stations by administrative region
Station Selection	Select and zoom to specific stations
Distance Measurement	Measure distances and areas
Mobile Support	Works on smartphones and tablets
PWA Installation	Install as a standalone app on Android

Table 4: Prescriptive Decision Matrix for Achieving Sub-centimeter RMS in Static GNSS Surveys

Baseline Length	Session Duration (Hours)	Ionospheric Activity	Sky Visibility (& PDOP)	Expected Horizontal RMS (cm)
< 5 km (Short)	0.5 – 1.0	Low / Moderate	Unobstructed (>15° cut-off)	0.3 – 0.6
	1.0 – 2.0	High (Scintillation)	Obstructed (Trees/Urban)	0.5 – 0.8
5 – 20 km (Medium)	1.0 – 2.0	Low	Unobstructed	0.4 – 0.7
	2.0 – 4.0	Moderate / High	Partially Obstructed	0.6 – 1.0
20 – 50 km (Long)	2.0 – 4.0	Low	Unobstructed	0.6 – 1.0
	4.0 – 8.0	Moderate / High	Unobstructed (Dual-freq req.)	0.8 – 1.5
> 50 km (Very Long)	4.0 – 6.0 (Static)	Low	Unobstructed	1.0 – 2.0
	6.0 – 12.0	High / Geomagnetic Storm	Unobstructed (Requires IGS orbits)	1.5 – 3.0

Note: Values assume dual-frequency GNSS receivers, 1 Hz sampling interval, and a fixed (non-ambiguity-float) baseline solution processed with scientific software (e.g., Bernese, GIPSY, or Leica Infinity). RMS refers to the coordinate repeatability in the horizontal plane post-network adjustment. For vertical RMS, multiply values by 1.5–2.0. Network-RTK or PPP-AR solutions can reduce the upper bounds by 30% if correction data are available. **Source:** NOAA/NGS technical memoranda on GNSS geodetic control surveys (e.g., NOAA TM NOS NGS 92), Static GNSS accuracy guides and manuals (e.g., CHCNAV’s “GNSS Static Measurements: Accuracy Guide”, Carleton GETSI static methods manual)

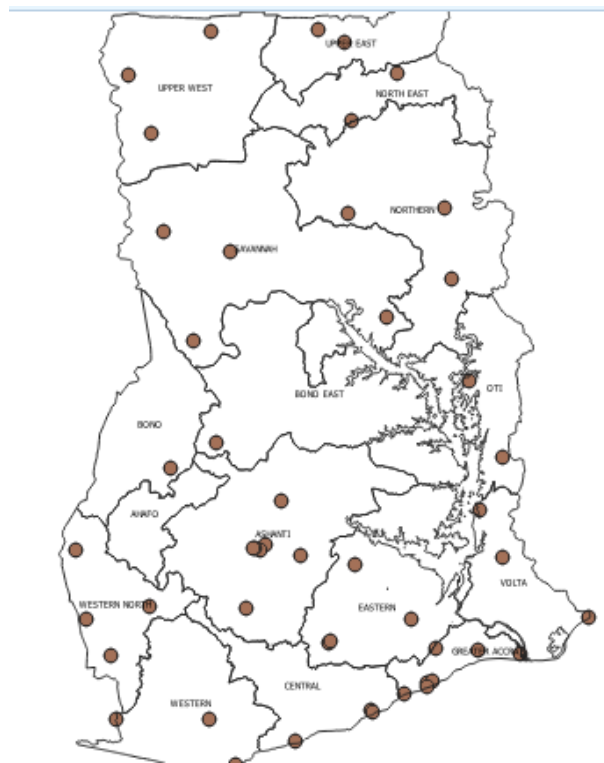
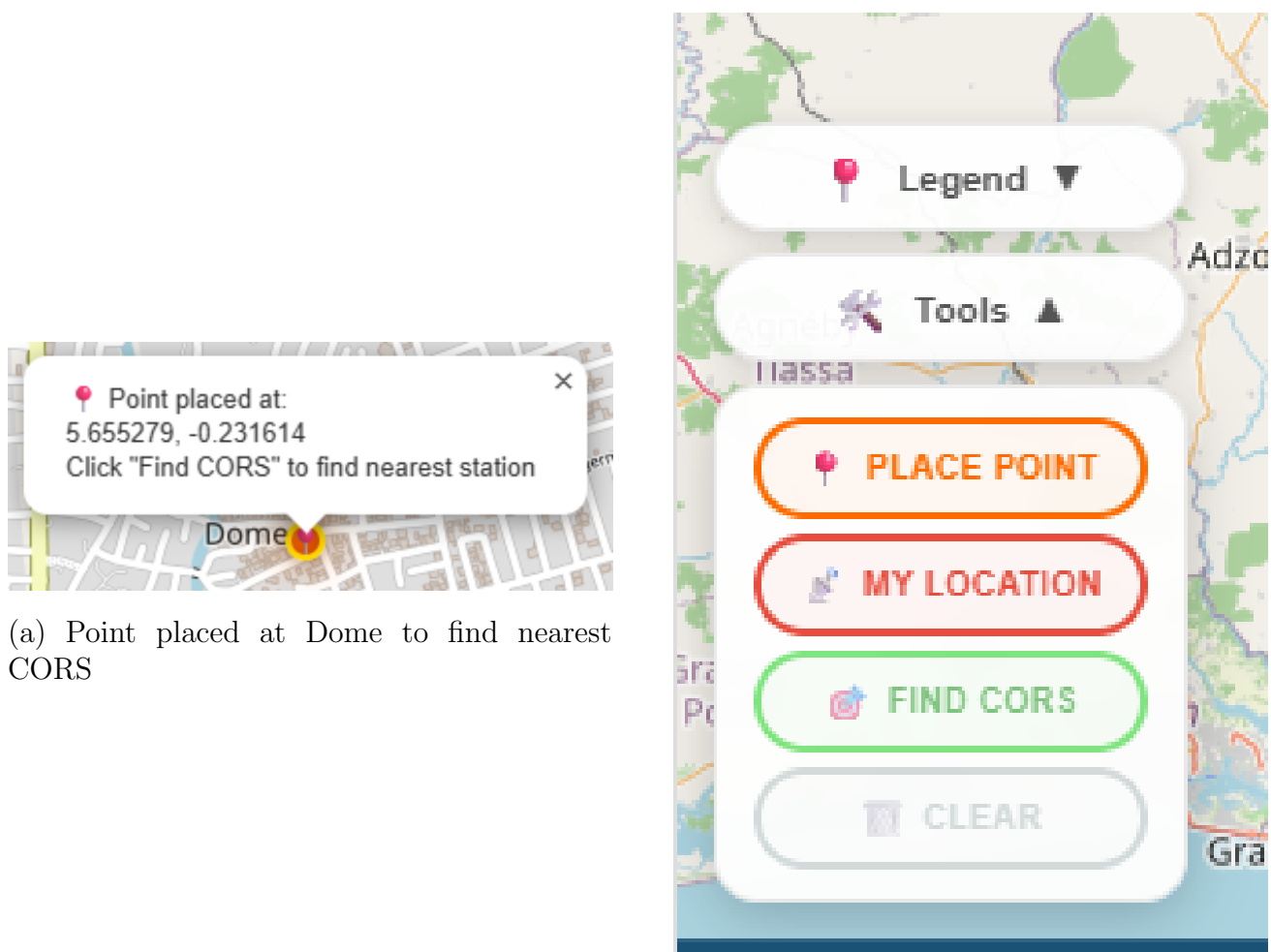


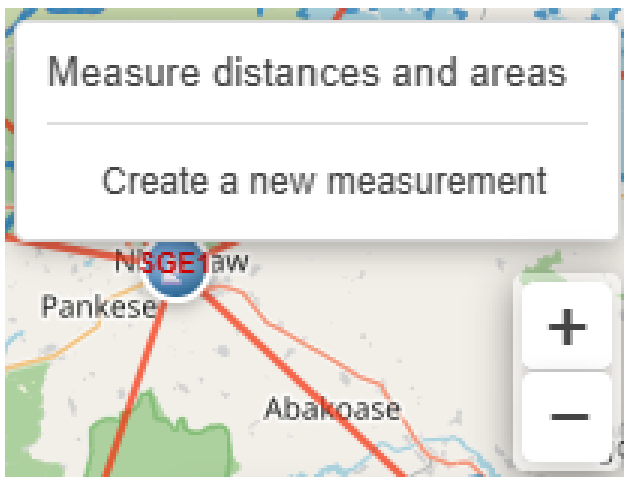
Figure 7: Spatial distribution of CORS on map of Ghana



(a) Point placed at Dome to find nearest CORS

(b) Nearest CORS found

Figure 8: Determining the nearest CORS from a placed point: (a) User places a point at Dome, (b) The application identifies and highlights the nearest CORS



(b) Nearest CORS found



(b) Nearest CORS found

Figure 9: Determining the nearest CORS from a placed point: (a) User places a point at Dome, (b) The application identifies and highlights the nearest CORS

Acknowledgments

The author would like to acknowledge:

- The Survey and Mapping Division (SMD) of Ghana Lands Commission
- The Licensed Surveyors Association of Ghana (LiSAG)
- GMX Systems Ghana Limited
- The open-source community behind QGIS [3] and Leaflet [4]

About the Author

Surv. Kwabena Boafo Matthew (MGhIS), known in international circles as Matthew is the Head of the Land Information Unit at the Lands Commission, where he leads the design, deployment, and management of geospatial and land information systems. He specializes in desktop and web-based application development, with a strong focus on building custom QGIS plugins and GIS/LIS tools that support land administration, cadastral mapping, and property verification workflows.

He serves as a trainer in GIS/LIS and database management, delivering practical instruction in spatial data capture, analysis, and database design using PostgreSQL, PostGIS, GeoServer and related technologies. His work bridges geomatics engineering and software development, integrating Python, Flask, and spatial databases into production systems for government and institutional use.

Kwabena has been actively involved in the development and implementation of both local and international land information systems, from requirements gathering and data modeling to system rollout and user training. He combines technical expertise in geospatial engineering with leadership experience in managing digitization projects, coordinating survey and mapping teams, and advancing digital transformation in land administration.

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